Position-Conditioned Deformation Prediction with Unknown Material and Environmental Interactions
Supplementary Material

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1. Physical model and units

The canonical block occupies $[0, 0.24] \times [-0.04, 0.04]^2$ metres. Density is $625\,\mathrm{kg/m^3}$, giving mass $0.96\,\mathrm{kg}$. Poisson ratio is 0.35. For local Young modulus $E(X)$, $\mu = E/[2(1+\nu)]$ and $\lambda = E\nu/[(1+\nu)(1-2\nu)]$. The material heterogeneity bases are $b_1 = \exp[-((X_x - 0.06)/0.07)^2]$ and $b_2 = \exp[-((X_x - 0.18)/0.07)^2]$.

Each end-face coupling has total stiffness K in N/m, with density normalized by its integrated footprint. For an end-face point (y, z),

$$w(y,z) = \left(1 - (y/0.04)^2\right)_+ \left(1 - (z/0.04)^2\right)_+ \times \exp\left[-\frac{(y-c_y)^2 + (z-c_z)^2}{2(0.03)^2}\right]. \quad (1)$$

The four possible footprint centers are $(0, -0.025)$, $(0, 0.025)$, $(-0.025, 0)$, and $(0.025, 0)$ metres. The three layout hypotheses pair the two lower footprints, lower and upper footprints, or opposite y footprints. Tool rotations act about the respective end-face centers. Attachment footprints move with the object/tool association; the distributed environmental potential is evaluated at current world positions.

2. Latent inference

The eleven latent coordinates comprise $\log E_0$, $\log K_l$, $\log K_r$, two gravity tilts, an environmental amplitude, material heterogeneity, three spatial-center offsets, and a width coordinate. Gravity is

$$g = 9.81 \frac{(z_3, z_4, -1)}{\|(z_3, z_4, -1)\|}, \quad (2)$$

with zero-based latent indexing. The Gaussian amplitude is $A = 0.1 z_5$ joules, $h = z_6$, the center is $(0.12, 0, -0.09) + (0.06 z_7, 0.04 z_8, 0.04 z_9)$ metres, and $s = 0.08 \exp(0.5 z_{10})$ metres.

Bounds are $E_0 \in [3, 100]\,\mathrm{kPa}$, $K \in [300, 60000]\,\mathrm{N/m}$, gravity tilts in $[-0.8, 0.8]$, $A \in [-0.5, 0.5]\,\mathrm{J}$, $h \in [-1.2, 1.2]$, center coordinates $z_{7:9} \in [-1, 1]$, and width coordinate $z_{10} \in [-1.2, 0.8]$. The weak Gaussian regularizer has means $(\log 20000, \log 5000, \log 5000, 0, \ldots, 0)$ and scales $(1, 1, 1, 0.3, 0.3, 2, 1, 0.6, 0.6, 0.6, 0.8)$.

We use bounded L-BFGS with at most 80 iterations, objective tolerance 10^{-9} and gradient tolerance 10^{-5}. A coarse solve screens all three layouts. Fine inference retains layouts within one normalized objective unit of the best coarse result. For post-acquisition inference, initialization chooses the lower acquired-data objective between the coarse post-acquisition fit and the previous fine fit. Neither initialization nor layout selection uses the held target response. Optimizer termination messages are retained separately from predictive error.

3. Reference and gradient checks

The reference uses quadratic hexahedral finite elements with Newton equilibrium solves and spatial refinement. The independent inverse backend assembles a dense tangent in double precision. Tests compare energy, residual and tangent with the sparse implementation and compare implicit gradients with finite differences. Further checks cover footprint normalization, positive Jacobians, mesh interpolation, probe-target exclusion and whole-object translation symmetry.

Each accepted reference action must satisfy adjacent-grid and quadrature displacement RMS changes below $20\,\mu$m, residual below 10^{-9} N, force balance below 10^{-7} N and minimum sampled $J > 0.1$. These are observed refinement checks at the declared query points. Surface quadrature uses order six; an independent development comparison against order ten changed predicted displacements by less than one nanometre. The mesh has $4n \times n \times n$ quadratic elements and graded thickness coordinates. Full-field warm starts are interpolated using the actual graded mesh.

The two-resolution predictor is $\hat{u} = (9u_3 - 4Pu_2)/5$. It is an extrapolated displacement field rather than an exact equilibrium of one mesh. Its derivatives combine the two implicit equilibrium derivatives. Calibration accounts empirically for residual prediction discrepancy.

At the true parameters of development object 3, the

$n = 2$ predictor had coordinate errors of 0.271, 0.230 and 0.302 mm for nominal pose, stretch and bending. The $n = 3$ errors were 0.126, 0.103 and 0.138 mm, and the selected two-resolution predictor gave 0.079, 0.068 and 0.084 mm. A direct $n = 4$ solve gave 0.059, 0.047 and 0.064 mm. This diagnostic motivated the selected accuracy–cost trade-off before held-out fitting; it is a forward-model check with privileged parameters, separate from inverse prediction accuracy.

All 87 primary and spatial-transfer reference fields passed these checks. Thirty-nine used resolution $n = 6$, forty-four used $n = 8$, and four used $n = 10$. The largest accepted action-level spatial RMS change was $19.35\,\mu$m; the largest quadrature RMS change was $0.278\,\mu$m. The minimum sampled deformation Jacobian was 0.727, and the largest reaction imbalance was 1.91×10^{-11} N. These counts concern independently refined reference objects; the separate 512-object neural training bank uses the two-resolution predictor.

4. Observation interface and action splits

All predictors receive three noisy visible position sets and prescribed tool poses. Marker correspondence and camera calibration are known. The front face provides 27 points; the 81-point evaluation set additionally includes 47 hidden surface points and seven interior points. Depth noise has standard deviation 0.3 mm and rasterization introduces quantization. The common positional fitting scale is 0.3 mm per coordinate.

The acquired prefix is rest, right-tool $+6$ mm y, and right-tool $+6$ mm z. Held local actions are right-tool $+3$ mm y, symmetric 3 mm stretching, opposed 6 mm y motion, opposed 0.15 rad twist, right-tool 0.15 rad bending, diagonal translation, and right-tool $+12$ mm y. The transfer test uses the first nine familiar-parameter test objects. The transfer challenge translates both tools by ± 40 mm along x or y, $+40$ mm along z, or -25 mm along z. The -40 mm y response remains held out. The other five translations are probe candidates. The fixed policy uses $+40$ mm y; random selection uses independent seed ranges $292000 + i$, $293000 + i$ and $294000 + i$ for development, calibration and test object i, respectively.

Calibration uses twelve separate objects. The pre-acquisition scale and each policy's post-acquisition scale are fitted independently on those objects. With twelve object scores and a 90% object target, the finite-sample rank is thirteen times 0.9 rounded up, or twelve: the largest calibration score. This intentionally exposes the data requirement for reliable object-level uncertainty. Attachment-layout counts are fixed and balanced within each cohort. We use the order statistic for empirical calibration and report measured coverage; the general exchangeability result is not asserted for this stratified design or for OOD shifts.

5. Baseline implementations

PhysTwin uses the official spring-mass simulation and per-spring Adam optimizer, with our observation and positional-control adapter. The adapter uses 81 object mass points at the shared canonical query locations and 18 prescribed controller points, with volume-weighted object masses. Only the last second of each five-second action contributes observed settled positions; intermediate target shapes are not fabricated. The one-second tool ramp and four-second hold use a 0.2 ms time step. Development selection tests log-stiffness regularization strengths 0, 0.01 and 0.1. The selected strength is 0.1, with initial spring stiffness 3 and 100 epochs. The regularizer shrinks bulk spring log stiffness toward the mean in three material-coordinate bins, while controller springs remain free. This regularizer is our adaptation. A fixed-weight timestep-halving check changed predictions by less than $0.5\,\mu$m on the development case.

PhysGaussian uses its official fixed-corotated MPM core, with our compliant positional coupling and inverse-fitting adapter. A normalized 6 mm attachment layer, APIC transfer, a one-second ramp and eight-second simulation are selected on development checks. The fine configuration uses 14,161 particles, grid resolution 64 and a 0.1 ms step; the coarse fit uses grid 32 and 0.2 ms. The adapter fits elasticity, heterogeneity, coupling and gravity direction from positions. Development checks found that APIC reduced twisting hold-duration sensitivity relative to PIC; the numerical transfer setting was selected before held-out fitting. Timestep-halving changes are reported separately from physical-model error. The reported comparison concerns these adapted physical components; it does not evaluate the authors' complete reconstruction or appearance pipelines.

At the final fitted parameters of development object 1, extending the selected MPM run from 8 to 16 seconds changed coordinate RMS displacement by 0.00482, 0.000032 and 0.000031 mm for nominal pose, stretch and twist. Halving the timestep from 0.1 to 0.05 ms at eight seconds changed those predictions by 0.0322, 0.0354 and 0.0283 mm. These fixed-parameter checks were completed after the method freeze and did not change its settings. They quantify settling and timestep sensitivity separately from constitutive and inverse-fitting error.

Neural baselines use 512 disjoint synthetic objects, split by object into training and validation. Their targets use the same two-resolution forward predictor as the proposed inverse method. A point-query MLP and a rank-32 spatial-basis predictor are trained for 20,000 AdamW updates. The spatial basis uses only training objects. Architecture selection uses validation error alone, separately for the ID-action and all-action training variants. Neither architecture receives physical parameter labels. Both variants remain

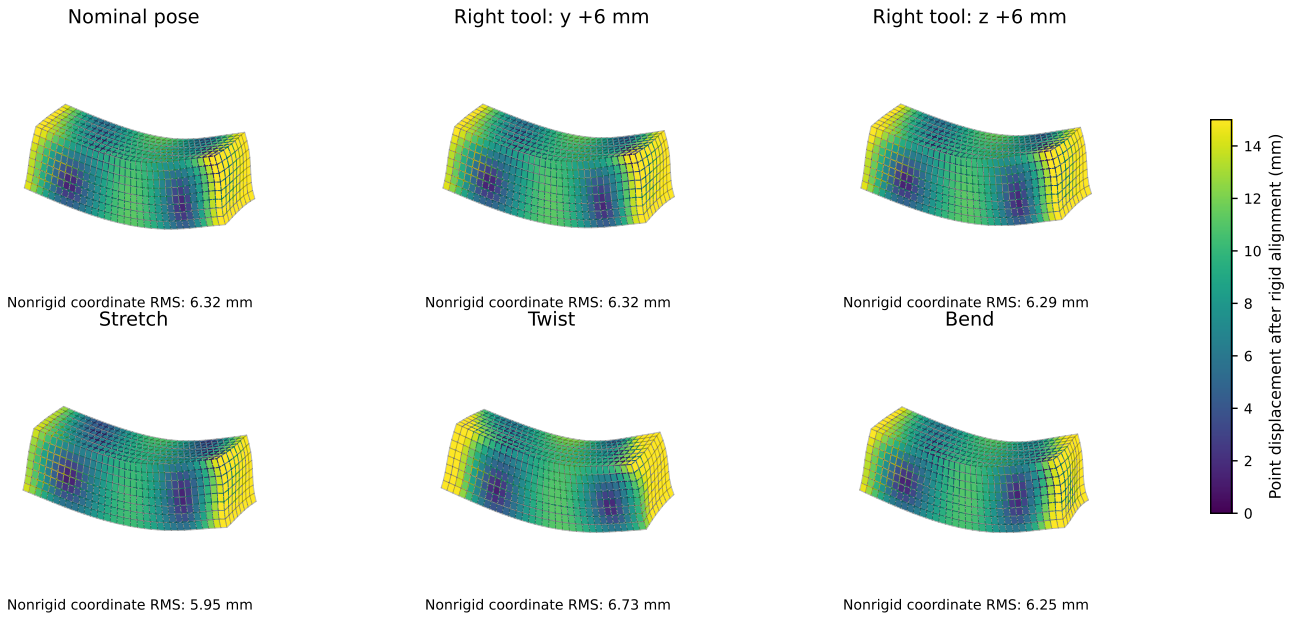


Figure 1. Development reference geometry at actual deformation scale. The top row supplies the three observed configurations; the bottom row illustrates new positional interactions. Color measures displacement remaining after rigid alignment. The block has millimetre-scale nonrigid deformation, with no visual amplification.

separately named in the results.

The selected ID-action and all-action neural training loops took 3.02 and 3.09 minutes, respectively, on the shared workstation. These recorded times exclude training-bank generation, data loading and basis construction. The common bank contains twelve paired-resolution positional queries per object; its simulation cost is separate from network training and per-object prediction. All runtime measurements reflect concurrent workstation use rather than isolated hardware benchmarks.

The accompanying compute report separately accounts for post-acquisition optimization. It counts each distinct acquired action once per object, records reuse when fixed, random and adaptive policies choose the same action, and separates coarse screening from fine fitting and CPU from GPU execution. Recorded fit times sum optimizer wall times across screened layouts; they exclude lock waiting, prediction, uncertainty calculation and reference generation. Calibration and held-test adaptation costs remain separate. These measurements describe the executed computation, rather than the latency of a dedicated deployment.

All 394 declared primary jobs completed. Median recorded job time for the 18 PhysTwin adaptations was 19.5 minutes, versus 239.2 minutes for the 18 MPM adaptations (ranges 14.8–24.9 and 111.5–398.7 minutes). The 54 primary fine-inference jobs took median 59.3 minutes on CPU (30 jobs) and 37.6 minutes on GPU (24 jobs). These are heterogeneous, concurrently executed jobs; timing includes each job's declared work and is not a controlled hardware comparison.

Optimizer termination is recorded separately from prediction quality. Of the 54 primary fine-inference jobs, 53 reached the frozen iteration budget and one met the relative-objective stopping criterion. Of the 18 MPM fits, 17 met the parameter-step criterion and one reached its function-evaluation budget. PhysTwin used its fixed 100-epoch schedule. Evaluation uses the resulting frozen-budget predictions without extending individual test-case budgets. Numerical reference qualification concerns the forward equilibrium solutions, not global convergence of these inverse optimizers.

The completed acquisition study contains 94 distinct fitting stages: 44 for calibration, 32 for held-test spatial inference and 18 for the matched material-indexed ablation. Across 47 coarse CPU stages, median recorded optimizer time is 6.7 minutes. The 22 fine CPU stages have median 83.9 minutes (range 39.9–123.5), and the 25 fine GPU stages have median 62.6 minutes (range 15.2–148.5). Each stage includes its screened layouts. These counts deduplicate shared policy actions. Different objects, layout counts and concurrent loads prevent interpreting the CPU/GPU figures as a controlled speed comparison; the compact acquisition compute report retains every stage and its source hash.

The selected spatial-basis models had validation coordinate RMSE of $0.116\,\mathrm{mm}$ for ID-action training and $0.124\,\mathrm{mm}$ for all-action training, compared with 0.207 and

0.237 mm for the MLP. In the PhysTwin development selection, log-stiffness regularization strengths 0, 0.01 and 0.1 gave coordinate RMSE pooled over held actions of 1.048, 0.816 and 0.710 mm on development object 1. Selection therefore used predictive validation, rather than the smallest acquired-position residual. Final cohort tables instead average action-level RMSE within each object, as defined in the main protocol.

Completed neural baseline evaluation. Across the full familiar-parameter, material-shift and environmental-shift cohorts, the all-action model has new-local-motion coordinate RMSE of 0.119, 0.402 and 0.216 mm, respectively. The ID-action model has 1.528, 1.551 and 1.488 mm. These are equal-weight object means over the five declared local interaction patterns. The all-action variant has seen these local action families on disjoint training objects; their novelty applies to the ID-action variant. On the six whole-object translations of all nine transfer objects, errors are 11.436 mm for all-action training and 16.514 mm for ID-action training. Neither training variant includes these large global translations. Both variants are retained in the comparison; test results do not select between them. The accompanying report includes per-object values, paired bootstrap intervals, incremental-response errors and source hashes.

6. Separately declared unseen force laws

The primary environment-shift cohort changes parameters within the single-Gaussian potential family. A separate six-object extension tests two environmental force laws outside that family while retaining unknown material coefficients, gravity direction and tool coupling. Its protocol was declared after the primary method freeze and before generating any extension reference fields or fitting its objects. All predictors retain their frozen settings and learned weights.

Three objects use an anisotropic harmonic potential,

$$\phi_h(x) = \frac{(x-c)^\top Q(x-c)}{2|\Omega_0|}, \qquad Q = R\,\mathrm{diag}(10, 25, 40)\,R^\top, \tag{3}$$

where Q has units N/m, $c = (0.12, 0, -0.06)$ m, and each rotation vector defining R is sampled componentwise from $[-0.5, 0.5]$ radians. The remaining three objects use two spatial Gaussian components,

$$\phi_b(x) = \frac{1}{|\Omega_0|}\sum_{j=1}^{2} A_j \exp\left(-\frac{\|x-c_j\|^2}{2s_j^2}\right). \tag{4}$$

The centers are $(0.07, -0.035, -0.06)$ and $(0.17, 0.035, -0.09)$ m, with widths 0.045 and 0.055 m. Amplitudes are $+0.07$ and -0.09 J, each multiplied independently by a uniform factor in $[0.85, 1.15]$. These potentials generate a spatially varying restoring field and a signed two-lobed field, respectively.

Each force-law group contains one object per attachment layout. Elasticity, material heterogeneity, coupling stiffness and gravity orientation are sampled within the familiar ranges using seed 295001; independent observation noise uses seed $296000 + i$. Each object has the original ten local actions plus six whole-object translations. The same three prefix observations are supplied to every predictor. The spatial inverse, material-indexed-load ablation, both selected neural models, PhysTwin adaptation and PhysGaussian adaptation are compared on all six objects. The extension has no additional training or probe acquisition. Reference generation uses the original refinement and equilibrium thresholds, with separate energy-gradient, tangent and reaction-balance tests for both new fields. Its results remain separate from the 42-object primary benchmark and the nine-object acquisition experiment.

All six extension references passed qualification: three used $n = 6$ and three used $n = 8$. The largest action-level spatial RMS change was 18.77 μm and the largest quadrature RMS change was 0.0444 μm. The minimum sampled deformation Jacobian was 0.832. The six fields, observations and qualification records were hashed before extension inference began.

7. One-response acquisition results

Figure 2 compares prediction error, coverage and interval width after one acquired response, alongside the matched-frame ablation.

8. Reproducibility and provenance

A pre-evaluation SHA-256 manifest freezes the inference implementation, benchmark configuration, object cohorts, transfer protocol, neural weights and native simulator sources. Numerical fields and rendered data remain local; code, compact reports and manuscript sources are versioned. The study uses CPU workers and GPUs 0–2 on a four-GPU workstation, retaining the fourth GPU for existing desktop use. Raw prediction files precede separate scoring. Object-level bootstrap comparisons use paired objects, not individual coordinates as independent replicates. Explicit completion counts accompany every declared cohort.

9. Scope

This experiment isolates environmental and material ambiguity for known object geometry, synthetic correspondence and settled conservative mechanics. It evaluates unseen positional interaction patterns, material/environment parameter shifts, and two environmental potential families outside the fitted model. Extensions to impact, changing external

Method	Familiar parameters		Material shift		Environment shift	
	New motions	2× amplitude	New motions	2× amplitude	New motions	2× amplitude
Spatial inverse (ours)	0.026	0.054	0.021	0.023	0.031	0.050
Neural (ID actions)	1.555	0.082	1.493	0.222	1.505	0.156
Neural (all actions)	0.061	0.059	0.202	0.151	0.148	0.119
PhysTwin core adaptation	0.386	0.295	0.639	0.428	0.655	0.465
PhysGaussian MPM adaptation	0.105	0.159	0.117	0.116	0.208	0.155
Material-indexed load	0.023	0.048	0.032	0.069	0.036	0.094
Nominal environment	0.106	0.103	0.276	0.282	0.281	0.246

Table 1. Coordinate RMSE (mm), lower is better. Incremental response error after subtracting each method's nominal-pose prediction, on the same matched 18-object subset. Object-level bootstrap intervals and familiar-direction queries are provided in the accompanying report.

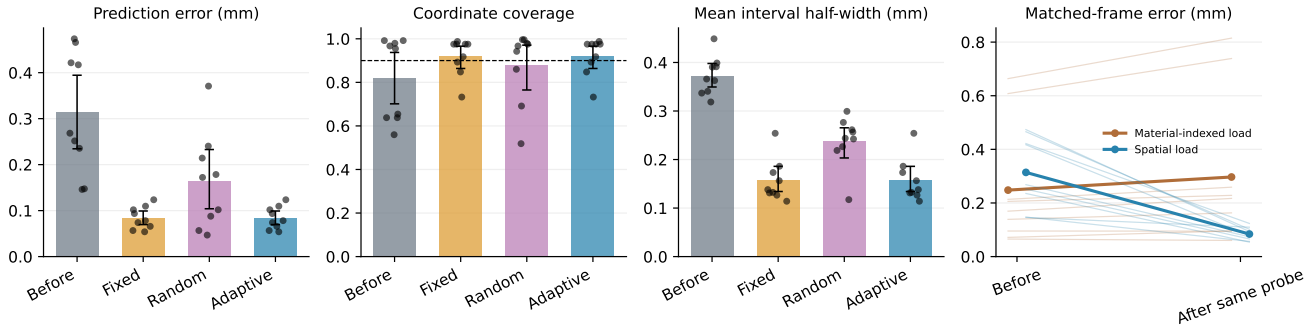


Figure 2. One-response adaptation on nine held objects. Error, predictive coverage and interval width are evaluated together, using independent calibration for each policy. Columns show means, points denote objects, and error bars show object-bootstrap intervals. The final panel compares spatial and material-indexed loading before and after the same acquired response.

Cohort	Coordinates	Objects	Half-width
Familiar parameters	98.3%	18/18	0.169
Material shift	95.9%	10/12	0.168
Environment shift	98.7%	12/12	0.181

Table 2. Prediction-interval coverage for held local actions. Coordinates is mean coordinate coverage; Objects counts cases with at least 90% of coordinates covered. Half-width is in mm. One fixed scale is fitted on twelve independent calibration objects.

forces, arbitrary topology and real visual correspondence require additional validated data and observation models.

LLM Usage Statement

OpenAI Codex assisted with research planning, implementation, experiment execution and analysis, related-work lookup, figure preparation, and manuscript drafting. The manuscript and its experimental claims require review and approval by the human authors before submission.